

CASTELO BRANCO, Portugal

WMO: 085700

Lat: 39.8394N

Lon: 7.4786W

Elev: 387

StdP: 96.76

Time Zone: 0.00 (EUW)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	2.4	-6.7	2.2	6.0	-4.5	2.7	7.2	9.3	10.4	8.2	9.9	2.5	30	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.2	36.6	19.3	35.0	18.9	33.5	18.5	20.7	32.1	20.1	31.2	19.5	30.1	2.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	13.4	21.4	17.1	12.8	21.1	16.3	12.2	21.0	61.0	32.0	58.9	31.3	57.0	30.2	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.7	6.7	5.9	DB	-1.3	39.2	1.4	1.5	-2.3	40.3	-3.1	41.2	-3.8	42.1	-4.8	43.2
			WB	-2.9	22.0	1.5	0.6	-4.0	22.4	-4.8	22.7	-5.7	23.1	-6.7	23.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	8.5	9.7	12.3	13.9	17.5	22.0	24.7	25.1	21.9	17.1	11.6	8.9
	DBStd	6.73	2.59	2.60	3.12	3.49	3.91	4.13	3.54	3.36	3.39	3.26	2.79	2.45
	HDD10.0	175	59	33	13	4	0	0	0	0	0	1	14	50
	HDD18.3	1511	304	241	188	139	65	12	1	1	7	62	201	292
	CDD10.0	2418	13	26	83	122	232	359	455	469	357	222	64	17
	CDD18.3	714	0	0	1	7	38	121	198	211	113	25	0	0
	CDH23.3	7958	0	0	7	64	400	1399	2374	2471	1066	176	0	0
	CDH26.7	3771	0	0	0	6	120	653	1238	1286	424	45	0	0
Wind	WSAvg	2.7	2.6	2.9	3.0	2.9	2.7	2.8	2.9	2.7	2.4	2.4	2.7	2.5
Precipitation	PrecAvg	664	81	60	55	58	52	14	6	7	40	110	93	89
	PrecMax	1056	385	167	226	151	157	52	30	34	127	214	298	307
	PrecMin	404	0	0	0	7	0	0	0	0	2	17	0	9
	PrecStd	189	77	42	55	37	46	15	8	8	34	59	77	82
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.5	19.6	24.3	27.5	32.6	37.2	38.4	38.6	35.3	30.9	21.4	16.8
		MCWB	11.4	11.7	13.4	16.5	18.1	19.7	19.7	19.6	18.8	16.1	14.7	12.3
	2%	DB	15.3	17.8	22.1	25.2	29.9	34.8	36.5	36.6	33.1	27.6	19.2	15.3
		MCWB	10.9	11.5	12.5	15.2	17.4	18.8	19.2	19.2	18.0	15.8	13.9	12.3
	5%	DB	13.9	16.1	20.4	23.2	27.5	32.9	34.9	34.9	31.1	25.1	17.5	14.1
		MCWB	10.8	10.8	12.2	14.6	16.4	18.5	19.0	18.7	17.5	15.5	13.1	11.6
	10%	DB	12.7	14.4	18.3	20.8	25.3	30.6	33.1	33.1	29.1	22.8	15.9	13.0
		MCWB	10.3	10.3	11.6	13.6	15.8	18.1	18.6	18.4	16.9	15.0	12.7	10.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	14.2	15.5	17.9	19.9	21.1	21.5	21.4	20.1	19.4	17.1	14.6
		MCDB	14.9	16.4	20.6	25.5	30.0	32.2	33.3	33.2	30.5	23.4	18.8	15.2
	2%	WB	12.7	13.0	14.1	16.3	18.5	20.2	20.6	20.6	19.3	18.3	15.5	13.4
		MCDB	13.8	15.1	18.5	23.4	26.9	31.0	32.6	32.2	28.6	21.9	17.0	14.3
	5%	WB	11.8	12.0	13.2	15.2	17.5	19.5	20.0	19.9	18.7	17.5	14.6	12.3
		MCDB	13.0	14.5	17.6	21.6	25.2	30.0	31.8	31.1	27.3	21.1	16.0	13.5
	10%	WB	10.8	11.0	12.4	14.1	16.5	18.7	19.3	19.3	18.0	16.6	13.6	11.3
		MCDB	12.1	13.7	16.7	19.4	23.6	28.7	30.5	30.2	25.9	20.2	15.2	12.5
Mean Daily Temperature Range	5% DB	MDBR	7.3	8.5	9.5	9.9	11.2	13.0	14.4	14.2	11.9	9.1	7.4	6.9
		MCDBR	8.8	10.7	12.3	12.9	13.9	14.9	15.4	15.2	14.1	12.6	9.0	8.1
		MCWBR	5.7	6.1	6.3	6.0	5.8	5.6	5.3	5.3	5.1	5.2	5.1	5.4
	5% WB	MCDBR	6.4	8.4	10.1	11.6	12.5	14.1	14.8	14.3	12.0	8.7	6.7	6.5
		MCWBR	4.9	5.5	5.9	5.9	5.7	5.5	5.5	5.5	4.8	4.6	4.6	5.1
Clear-Sky Solar Irradiance	taub	0.305	0.320	0.351	0.352	0.360	0.362	0.348	0.363	0.353	0.343	0.316	0.304	
	taud	2.523	2.480	2.405	2.411	2.414	2.423	2.432	2.391	2.447	2.502	2.526	2.528	
	Ebn at Noon	869	902	904	922	918	912	922	900	890	863	846	841	
	Edh at Noon	75	89	107	113	116	115	113	114	101	87	74	69	
All-Sky Solar Radiation	RadAvg	2.14	3.11	4.18	5.28	6.51	7.46	7.81	6.88	5.24	3.44	2.33	1.90	
	RadStd	0.32	0.43	0.57	0.59	0.65	0.42	0.22	0.28	0.38	0.38	0.36	0.27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+57

Nomenclature: See separate page